

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	35.7	14.9	2.3	0.8	0.3	46
U3 - Toxicity prior Beta(0.15, 0.85)	36.4	13.5	1.7	0	0.1	48.3
U2 - Toxicity prior Beta(0.3, 0.7)	36.6	15.5	3.5	1.1	0.6	42.7
U3 - Toxicity prior Beta(0.3, 0.7)	39	14.5	2.5	0.2	0.1	43.7
U2 - Toxicity prior Beta(0.5, 0.5)	38.9	16.5	3.2	0.7	0.6	40.1
U3 - Toxicity prior Beta(0.5, 0.5)	40.3	15.1	2.7	0.2	0.2	41.5
U2 - Toxicity prior Beta(1, 1)	36.6	17.7	3.4	1	0.3	41
U3 - Toxicity prior Beta(1, 1)	40.3	14	3	0.2	0.1	42.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.3	6.4	2.6	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	14.7	6.1	2.4	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	13.7	6.5	2.9	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	14.2	6.2	2.7	1.6	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	13.4	6.6	3.3	1.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	13.9	6.3	3.1	1.8	0.7	
U2 - Toxicity prior Beta(1, 1)	13.5	6.5	3.2	1.8	0.7	
U3 - Toxicity prior Beta(1, 1)	14	6.2	3	1.8	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	3.3	0.7	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.1	0.7	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3.1	0.7	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3	0.7	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	2.8	0.7	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	2.7	0.7	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	1.1	2.3	0.7	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	1.1	2.2	0.7	0.5	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.3	19.2	37.6	12.3	2.7	0.9
U3 - Toxicity prior Beta(0.15, 0.85)	31.2	20.3	34.2	11.7	1.6	1
U2 - Toxicity prior Beta(0.3, 0.7)	25.5	19.1	36.6	14.9	3	0.9
U3 - Toxicity prior Beta(0.3, 0.7)	29.4	19.9	35.2	12.8	1.8	0.9
U2 - Toxicity prior Beta(0.5, 0.5)	23.5	18.5	39.1	15.6	2.7	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	29.7	18.3	35.4	13.7	2.1	0.8
U2 - Toxicity prior Beta(1, 1)	23.1	17.5	39	16.3	3.4	0.7
U3 - Toxicity prior Beta(1, 1)	28.3	17.7	37.8	13.4	2.2	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	7.2	8.9	3.6	1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	7.4	8.6	3.6	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	7	9	4.1	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	7.2	8.9	3.9	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	7	9.4	4.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	7.1	9.2	4	0.8	
U2 - Toxicity prior Beta(1, 1)	8.1	6.8	9.2	4.4	1.3	
U3 - Toxicity prior Beta(1, 1)	8.7	6.9	9.3	4.1	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.7	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.6	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.1	1.6	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	2.9	1.6	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	0.6	2.6	1.5	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	2.5	1.4	0.8	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10	47.8	25.8	12	2.1	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	12.5	53.6	20.8	9.2	1.5	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	9.9	46.4	25.6	12.9	3.1	2.1
U3 - Toxicity prior Beta(0.3, 0.7)	13.6	49.5	22.9	9.7	2.2	2.1
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	46.9	26.1	12.9	3.3	2
U3 - Toxicity prior Beta(0.5, 0.5)	12.5	49.4	23	10.8	2.3	2
U2 - Toxicity prior Beta(1, 1)	9.5	45.6	26.8	12.9	3.2	2
U3 - Toxicity prior Beta(1, 1)	11.8	50.9	22.6	10.3	2.3	2.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.6	11	7.1	3.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	6.8	11.9	6.6	3.5	1	
U2 - Toxicity prior Beta(0.3, 0.7)	6.3	10.7	7.3	4.1	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	11.5	7	3.7	1	
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	10.7	7.4	4.3	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	6.2	11.5	7	3.9	1.2	
U2 - Toxicity prior Beta(1, 1)	5.9	10.5	7.4	4.4	1.6	
U3 - Toxicity prior Beta(1, 1)	6.1	11.4	7	4	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	1.8	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.8	1.6	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.7	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.6	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.4	1.5	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.4	3.1	1.5	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	3	1.4	0.9	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.1	6.7	17.3	51.7	16	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	12.5	8.8	16	49.1	13.4	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	7.4	6.1	16.8	49.8	19.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	11.6	8	15.6	50.4	14.3	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	6.6	5.5	16.4	50.6	20.7	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	7.7	14.8	50.5	16.3	0.2
U2 - Toxicity prior Beta(1, 1)	6.8	5.1	14.8	51.7	21.4	0.2
U3 - Toxicity prior Beta(1, 1)	10.5	6.7	15.1	50.7	16.8	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	4.5	6.6	10.6	4.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.4	4.6	6.7	10.6	3.7	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.3	6.5	10.6	4.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	4.4	6.7	10.8	3.8	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	4.2	6.4	10.8	4.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	4.4	6.5	11	4	
U2 - Toxicity prior Beta(1, 1)	3.9	4.1	6.2	10.9	4.9	
U3 - Toxicity prior Beta(1, 1)	4	4.2	6.4	11.1	4.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	1.5	2.4	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.5	2.4	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.4	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.3	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.3	0.7	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.1	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.4	32	27.6	23.4	11.2	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	8.9	40.5	24	18.2	8	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	30.8	28.1	23.1	11.5	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	38.4	25.9	18.8	7.6	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	6	30.4	28	22.7	12.6	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	38.3	24.5	19.5	8.2	0.3
U2 - Toxicity prior Beta(1, 1)	6.1	31	27.2	23.3	11.9	0.5
U3 - Toxicity prior Beta(1, 1)	8.4	38.6	24.4	19.5	8.6	0.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.9	6.8	7.2	7.3	4.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4	7.7	7.2	7.1	4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	6.6	7.2	7.4	5	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	7.3	7.3	7.3	4.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.6	6.5	7.1	7.6	5.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	7.3	7.1	7.5	4.5	
U2 - Toxicity prior Beta(1, 1)	3.6	6.4	6.9	7.6	5.4	
U3 - Toxicity prior Beta(1, 1)	3.7	7.3	7	7.4	4.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	3.1	1.9	2.3	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.9	2.2	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	3	1.9	2.4	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	1.9	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.9	2.4	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	3	1.9	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.8	2.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.9	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	64.5	15.2	4.6	1.8	0.4	13.5
U3 - Toxicity prior Beta(0.15, 0.85)	67.2	14.7	3.6	1.1	0.4	13
U2 - Toxicity prior Beta(0.3, 0.7)	61.5	18	5.9	1.9	0.6	12.1
U3 - Toxicity prior Beta(0.3, 0.7)	65.3	16.2	4.4	1.5	0.5	12.1
U2 - Toxicity prior Beta(0.5, 0.5)	60.3	18.7	6.8	2.5	0.6	11.1
U3 - Toxicity prior Beta(0.5, 0.5)	63.9	17.8	4.5	1.8	0.5	11.5
U2 - Toxicity prior Beta(1, 1)	59.7	19.4	6.4	2.2	0.6	11.7
U3 - Toxicity prior Beta(1, 1)	64.6	17.6	4.2	1.5	0.6	11.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.4	6.2	2.5	1.3	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	18	6.1	2.3	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	16.7	6.5	2.8	1.4	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	6.4	2.5	1.3	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	16	6.7	3.3	1.6	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	16.5	6.7	2.9	1.4	0.4	
U2 - Toxicity prior Beta(1, 1)	16	6.7	3.1	1.6	0.6	
U3 - Toxicity prior Beta(1, 1)	16.7	6.6	2.8	1.5	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.4	0.7	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.4	0.6	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.4	0.7	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.4	0.6	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.3	0.7	0.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.6	0.3	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	1.8	0.6	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.7	1.8	0.6	0.3	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.1	46.2	26.4	10	3.4	0.9
U3 - Toxicity prior Beta(0.15, 0.85)	15.4	51.4	22.9	7.5	2	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	13.1	44.5	26.4	11.1	4.2	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	14.5	49.4	23.9	9.3	2.1	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	43.4	28.4	11.8	4.8	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	13.4	48.5	25.4	9.7	2.2	0.8
U2 - Toxicity prior Beta(1, 1)	10.9	43.7	28	12.6	4.1	0.7
U3 - Toxicity prior Beta(1, 1)	12.7	50.2	24.7	9.1	2.5	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.6	12.1	6.4	2.8	1	
U3 - Toxicity prior Beta(0.15, 0.85)	7.9	13.2	5.9	2.2	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	7.2	11.9	6.7	3	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	7.5	13	6.2	2.5	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	6.7	11.9	7	3.1	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	7.1	12.9	6.3	2.7	0.8	
U2 - Toxicity prior Beta(1, 1)	6.7	11.8	6.8	3.3	1.2	
U3 - Toxicity prior Beta(1, 1)	7	13	6.3	2.7	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.4	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.4	1.2	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.3	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.3	1.2	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.2	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.2	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.1	0.5	0.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	46.9	11.7	3.9	0.8	0.7	36
U3 - Toxicity prior Beta(0.15, 0.85)	49	11.3	2.7	1	0.3	35.7
U2 - Toxicity prior Beta(0.3, 0.7)	45.9	14.6	4.5	0.9	0.7	33.4
U3 - Toxicity prior Beta(0.3, 0.7)	47.3	14.6	2.5	0.8	0.4	34.4
U2 - Toxicity prior Beta(0.5, 0.5)	45.9	15.4	4.5	1	0.9	32.3
U3 - Toxicity prior Beta(0.5, 0.5)	47.2	15.2	3.8	0.5	0.6	32.7
U2 - Toxicity prior Beta(1, 1)	46.8	13.5	5.6	1.3	1	31.8
U3 - Toxicity prior Beta(1, 1)	47.4	13.9	4.6	1.1	0.4	32.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.4	6.2	2.4	1.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	15.8	6	2.2	1.2	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	15	6.3	2.7	1.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	15.3	6.1	2.5	1.4	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	14.5	6.4	3.1	1.6	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	14.8	6.3	3	1.5	0.5	
U2 - Toxicity prior Beta(1, 1)	14.4	6.4	3.1	1.6	0.6	
U3 - Toxicity prior Beta(1, 1)	14.8	6.1	2.9	1.6	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	2.8	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	2.7	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	2.7	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1	2.6	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.4	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.4	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	1.9	0.6	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.9	1.9	0.6	0.3	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.8	13.8	19.7	16.8	28.1	3.8
U3 - Toxicity prior Beta(0.15, 0.85)	22.9	15.7	19.9	13.2	23.8	4.5
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	13.2	19.8	17.7	30.5	3.3
U3 - Toxicity prior Beta(0.3, 0.7)	21.7	15.8	19.7	14.8	24.5	3.5
U2 - Toxicity prior Beta(0.5, 0.5)	14.9	13	18.3	15.8	34.6	3.4
U3 - Toxicity prior Beta(0.5, 0.5)	19.9	14.9	19.3	14.9	27.7	3.3
U2 - Toxicity prior Beta(1, 1)	14	11.1	20	15.2	36.7	3
U3 - Toxicity prior Beta(1, 1)	20.3	14.5	18.6	13.3	29.9	3.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7	6	5.9	5.3	5.5	
U3 - Toxicity prior Beta(0.15, 0.85)	7.6	5.9	5.8	5.3	5.1	
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	5.8	5.8	5.5	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	7.1	5.7	5.7	5.7	5.5	
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	5.6	5.9	5.7	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	6.7	5.5	5.8	5.7	6	
U2 - Toxicity prior Beta(1, 1)	6.1	5.5	5.7	5.7	6.7	
U3 - Toxicity prior Beta(1, 1)	6.7	5.5	5.7	5.7	6.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1.5	1.8	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.5	1.5	1.8	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.4	1.5	1.8	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.3	1.5	1.8	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	1.4	1.9	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.1	1.4	1.8	0.9	
U2 - Toxicity prior Beta(1, 1)	0.6	2.8	1.5	1.7	0.8	
U3 - Toxicity prior Beta(1, 1)	0.7	2.8	1.5	1.6	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.9	40.2	16.2	6.5	1.8	19.4
U3 - Toxicity prior Beta(0.15, 0.85)	16.2	42.8	14.3	5.7	1.1	19.9
U2 - Toxicity prior Beta(0.3, 0.7)	13.4	41.6	17	7.8	2.6	17.6
U3 - Toxicity prior Beta(0.3, 0.7)	15.2	42.5	15.5	6.6	1.7	18.5
U2 - Toxicity prior Beta(0.5, 0.5)	12.4	41.4	19.5	8.4	2.6	15.7
U3 - Toxicity prior Beta(0.5, 0.5)	13.8	45.7	15.3	7.6	1.8	15.8
U2 - Toxicity prior Beta(1, 1)	11.7	40	18.9	9.9	2.8	16.7
U3 - Toxicity prior Beta(1, 1)	12.4	45.4	16.3	7.6	1.6	16.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	10.4	4.8	2.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	10	11	4.4	2.1	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	9.3	10.4	5	2.6	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	10.7	4.9	2.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.5	10.5	5.7	2.7	1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	11	5.3	2.6	0.8	
U2 - Toxicity prior Beta(1, 1)	8.6	10	5.6	3	1.1	
U3 - Toxicity prior Beta(1, 1)	8.7	10.8	5.2	2.7	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.9	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.7	1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.8	0.9	0.5	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	64.3	23.4	6.5	0.2	0	5.6
U3 - Toxicity prior Beta(0.15, 0.85)	70.4	20.7	3.4	0	0	5.5
U2 - Toxicity prior Beta(0.3, 0.7)	63.3	26.3	5.1	0.5	0	4.8
U3 - Toxicity prior Beta(0.3, 0.7)	67.5	23.2	4.1	0.5	0.1	4.6
U2 - Toxicity prior Beta(0.5, 0.5)	63.5	26.3	5	0.8	0	4.4
U3 - Toxicity prior Beta(0.5, 0.5)	67.7	23.8	3.6	0.5	0.2	4.2
U2 - Toxicity prior Beta(1, 1)	64	24.2	6.1	1.1	0.1	4.5
U3 - Toxicity prior Beta(1, 1)	67	24.4	3.4	0.7	0.1	4.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.1	7.9	3.5	1.4	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	16.9	7.6	3.2	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	15.3	8.1	3.9	1.6	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	16	7.9	3.6	1.6	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	14.7	8.3	4.2	1.8	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	15.4	8.1	3.8	1.7	0.4	
U2 - Toxicity prior Beta(1, 1)	14.9	8.1	4.1	1.8	0.4	
U3 - Toxicity prior Beta(1, 1)	15.5	8	3.8	1.8	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.4	1	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.3	0.9	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3.4	0.9	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3.3	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	0.9	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3	0.8	0.4	0	
U2 - Toxicity prior Beta(1, 1)	1.1	2.5	0.8	0.4	0	
U3 - Toxicity prior Beta(1, 1)	1.1	2.4	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.1	11.1	10.5	40.8	22.2	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	19.6	13.4	8.9	38.1	19.7	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	14.8	10.7	9.4	40.4	24.5	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	19.3	13.7	7.4	38.2	21.1	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	12.5	10.9	9.8	40.2	26.3	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	12.6	8	37.1	24.6	0.3
U2 - Toxicity prior Beta(1, 1)	11.9	11.1	8.4	39.4	28.9	0.3
U3 - Toxicity prior Beta(1, 1)	16.7	12.3	6.9	37.4	26.5	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	5.1	5.5	9.1	5.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.9	5.2	5.3	9.4	5.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	4.8	5.4	9.3	6.1	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	5	5.2	9.5	5.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	4.7	5.2	9.5	6.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	4.8	5	9.7	6	
U2 - Toxicity prior Beta(1, 1)	4.1	4.6	5	9.4	6.8	
U3 - Toxicity prior Beta(1, 1)	4.3	4.7	4.9	9.6	6.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.5	2.3	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.5	2.3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3	1.5	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.5	2.3	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.5	2.3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.4	2.2	1.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.5	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.5	2.1	1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	40.4	16	6.6	1.7	16.7
U3 - Toxicity prior Beta(0.15, 0.85)	21.2	40.8	14.2	5.6	1.5	16.7
U2 - Toxicity prior Beta(0.3, 0.7)	16.5	39.5	19.8	7.4	2.5	14.3
U3 - Toxicity prior Beta(0.3, 0.7)	19.1	42.9	15.4	5.8	2.1	14.7
U2 - Toxicity prior Beta(0.5, 0.5)	16.3	41.2	20.8	6.3	3.2	12.2
U3 - Toxicity prior Beta(0.5, 0.5)	19.6	41.8	17.7	6.2	2.1	12.6
U2 - Toxicity prior Beta(1, 1)	15.2	42.6	19.6	7.3	2.6	12.7
U3 - Toxicity prior Beta(1, 1)	18.2	43.6	17.5	5.8	2.2	12.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	9.7	5	3.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	10	9.9	4.8	2.9	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	9	9.5	5.5	3.3	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.3	9.8	5.1	3.2	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	9.5	5.8	3.6	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	9.8	5.5	3.5	1.3	
U2 - Toxicity prior Beta(1, 1)	8.3	9.3	5.8	3.7	1.6	
U3 - Toxicity prior Beta(1, 1)	8.6	9.8	5.6	3.5	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.1	1.2	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.1	1.2	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.2	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.1	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1.2	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1.2	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	2.9	1.1	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	2.9	1.1	0.7	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22	16.7	15.8	29.8	14.9	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	31.3	19	12.4	26.8	9.7	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	20.8	15.7	14.7	32	16.1	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	30.4	18	12	27.3	11.6	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	19.8	16.3	13.6	32.1	17.7	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	29.5	17.4	12	27.4	13.2	0.5
U2 - Toxicity prior Beta(1, 1)	18	16.8	12.5	32.2	19.8	0.7
U3 - Toxicity prior Beta(1, 1)	27.9	17.2	11	28.4	14.9	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6	6	5.7	7.6	4.7	
U3 - Toxicity prior Beta(0.15, 0.85)	6.6	6.1	5.5	7.5	4.3	
U2 - Toxicity prior Beta(0.3, 0.7)	5.6	5.7	5.6	7.8	5.1	
U3 - Toxicity prior Beta(0.3, 0.7)	6.2	5.8	5.4	7.8	4.7	
U2 - Toxicity prior Beta(0.5, 0.5)	5.3	5.6	5.5	8	5.5	
U3 - Toxicity prior Beta(0.5, 0.5)	5.7	5.7	5.3	8	5.1	
U2 - Toxicity prior Beta(1, 1)	5.2	5.4	5.5	8.1	5.7	
U3 - Toxicity prior Beta(1, 1)	5.7	5.5	5.2	8.2	5.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1.7	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.5	1.6	2.2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.4	1.6	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.4	1.5	2.2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.5	2.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.2	1.5	2.2	0.9	
U2 - Toxicity prior Beta(1, 1)	0.6	3	1.5	2.1	0.8	
U3 - Toxicity prior Beta(1, 1)	0.6	3	1.5	2.1	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	51.5	29	12.6	4.3	0.8	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	56.2	28.4	10.4	2.7	0.5	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	51.3	29.1	12.8	4.4	1	1.4
U3 - Toxicity prior Beta(0.3, 0.7)	58.2	26.6	10.2	3	0.4	1.6
U2 - Toxicity prior Beta(0.5, 0.5)	50.7	30.1	11.7	4.9	1.2	1.4
U3 - Toxicity prior Beta(0.5, 0.5)	57.7	27.4	10.4	2.7	0.4	1.4
U2 - Toxicity prior Beta(1, 1)	49.5	30.8	12.5	5	0.9	1.3
U3 - Toxicity prior Beta(1, 1)	56.3	28.7	10.2	3.4	0.3	1.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	8.3	5.9	4.2	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	10.7	8.2	5.4	4	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	8.2	6	4.5	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	8	5.6	4.3	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	8.1	6.1	4.7	2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	7.9	5.7	4.5	1.9	
U2 - Toxicity prior Beta(1, 1)	8.8	8	6.2	4.8	2.1	
U3 - Toxicity prior Beta(1, 1)	9.7	7.9	5.7	4.6	2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	4.1	1.8	1.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.3	4	1.7	1.5	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	4	1.8	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.3	3.9	1.7	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.8	1.7	1.6	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.7	1.6	1.5	0.4	
U2 - Toxicity prior Beta(1, 1)	1.1	3.6	1.7	1.5	0.3	
U3 - Toxicity prior Beta(1, 1)	1.3	3.6	1.6	1.5	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	40.4	24.1	16.6	11.7	7.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	45.5	26	14.7	9.4	4.4	0
U2 - Toxicity prior Beta(0.3, 0.7)	39.4	24.1	16.5	12.1	7.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	45.2	24.1	15.5	10.8	4.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	39.8	24.3	17.2	10.9	7.8	0
U3 - Toxicity prior Beta(0.5, 0.5)	45.3	24.6	15.2	9.8	5.1	0
U2 - Toxicity prior Beta(1, 1)	39.5	24.8	17.2	11.5	7	0
U3 - Toxicity prior Beta(1, 1)	45	25.5	15.4	9.8	4.3	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	4.8	5.9	8.2	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	5.2	6.2	8	5.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	4.9	6	8.1	6.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	5.1	6.2	8.2	6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	4.7	6	8.2	6.8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	5.1	6.2	8.1	6	
U2 - Toxicity prior Beta(1, 1)	4.3	4.7	6	8.2	6.8	
U3 - Toxicity prior Beta(1, 1)	4.5	5.1	6.2	8.1	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	2	2.9	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2.2	2.9	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	2	2.9	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	2.2	2.9	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	2	3	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	2.2	2.9	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	2.1	2.9	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	2.2	2.9	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.9	57.7	17.9	2.4	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	25	59.5	14	1.4	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	20.1	59.4	17.3	3	0.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	23.2	60.3	14.5	1.9	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	19.6	56.4	20.5	3.3	0.1	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	21.7	59.1	17.1	2	0	0.1
U2 - Toxicity prior Beta(1, 1)	19.3	57.2	20	3.2	0.1	0.2
U3 - Toxicity prior Beta(1, 1)	20.9	60.1	16.6	2.2	0.1	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	13.9	5.6	1.2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	14.4	5	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.7	14	5.9	1.3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	14.6	5.1	1.1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	13.9	6.1	1.6	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	14.4	5.4	1.2	0.1	
U2 - Toxicity prior Beta(1, 1)	8.2	13.8	6.1	1.7	0.2	
U3 - Toxicity prior Beta(1, 1)	8.7	14.5	5.4	1.2	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.8	1.3	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.8	1.1	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.3	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.1	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.2	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.1	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.4	3.2	1.1	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	1	0.3	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12	8.8	10	15.4	53.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	19.2	12	9.5	13.8	45.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	11.6	8.7	9.9	14.2	55.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	18.5	10.9	10	12.5	48.1	0
U2 - Toxicity prior Beta(0.5, 0.5)	13	7.2	9.8	13.1	56.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	18.6	10.6	9.9	12	48.9	0
U2 - Toxicity prior Beta(1, 1)	12.2	7.9	8.6	14.3	57	0
U3 - Toxicity prior Beta(1, 1)	17.3	9.6	9.5	12	51.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	3.8	5.2	6.7	10.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.1	5.2	6.8	10.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	3.8	5.1	6.7	11	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4	5.1	6.8	10.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.8	5	6.6	11.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	3.9	5.1	6.8	10.7	
U2 - Toxicity prior Beta(1, 1)	3.3	3.7	4.9	6.6	11.5	
U3 - Toxicity prior Beta(1, 1)	3.5	3.8	5	6.8	11	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.5	2.5	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.5	1.5	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.5	1.5	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.5	1.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.7	57.8	19.9	2.1	0	2.5
U3 - Toxicity prior Beta(0.15, 0.85)	22.9	57.1	15.8	1.7	0	2.5
U2 - Toxicity prior Beta(0.3, 0.7)	17.1	57.4	20.5	2.5	0.1	2.4
U3 - Toxicity prior Beta(0.3, 0.7)	19.7	59.3	16.6	1.9	0	2.5
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	57.3	19.3	2.9	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	21	57.1	17	2.5	0	2.4
U2 - Toxicity prior Beta(1, 1)	17.7	55.4	21.8	3.1	0.1	1.9
U3 - Toxicity prior Beta(1, 1)	19.9	58.1	17.7	2.4	0	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.9	12.3	6.8	2.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.3	12.5	6.3	2.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	12	7.1	2.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	7.9	12.3	6.7	2.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	12	7.2	2.9	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	7.5	12.2	6.9	2.7	0.6	
U2 - Toxicity prior Beta(1, 1)	7.2	11.8	7.2	3	0.7	
U3 - Toxicity prior Beta(1, 1)	7.4	12.3	6.9	2.8	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.5	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.5	1.6	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.4	1.6	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.4	1.6	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.4	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.6	1.3	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	3.6	1.3	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.9	47.3	28.3	11.6	1.5	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	15.4	50.7	23.1	9.6	0.8	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	11	46.9	27.9	11.8	2	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	17	49.8	23.8	8.1	0.9	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	13	44.6	29	10.4	2.6	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	17.3	47.2	24.7	8.8	1.6	0.4
U2 - Toxicity prior Beta(1, 1)	11.9	42.6	28.8	13.6	2.6	0.5
U3 - Toxicity prior Beta(1, 1)	17.8	46.4	24	9.8	1.4	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.8	9.6	8.2	5.4	2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	10.2	7.8	5	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	9.3	8.3	5.6	2.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.9	9.9	7.9	5.3	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.4	9	8.1	5.9	2.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	9.6	7.9	5.6	2.1	
U2 - Toxicity prior Beta(1, 1)	4.4	8.7	8.1	6.1	2.7	
U3 - Toxicity prior Beta(1, 1)	4.7	9.4	7.8	5.7	2.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	2.3	1.7	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	2.2	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	2.2	1.8	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	2.1	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.6	2.1	1.8	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.6	2.1	1.7	0.5	
U2 - Toxicity prior Beta(1, 1)	0.4	3.4	2	1.7	0.4	
U3 - Toxicity prior Beta(1, 1)	0.5	3.5	2	1.6	0.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	56.7	1.8	0	0	0	41.5
U3 - Toxicity prior Beta(0.15, 0.85)	56.7	1.9	0	0	0	41.4
U2 - Toxicity prior Beta(0.3, 0.7)	57.3	2.3	0	0	0	40.4
U3 - Toxicity prior Beta(0.3, 0.7)	56.4	2.4	0	0	0	41.2
U2 - Toxicity prior Beta(0.5, 0.5)	58.3	2	0.1	0	0	39.6
U3 - Toxicity prior Beta(0.5, 0.5)	57.7	2.1	0.1	0	0	40.1
U2 - Toxicity prior Beta(1, 1)	57.8	2.1	0	0	0	40.1
U3 - Toxicity prior Beta(1, 1)	57.6	2.1	0.1	0	0	40.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.9	2.8	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20	2.7	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	19.9	2.9	0.3	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	19.8	2.9	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	19.8	3.1	0.4	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20	3	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	19.8	3	0.4	0	0	
U3 - Toxicity prior Beta(1, 1)	20	3	0.3	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.1	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.1	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	0.9	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	0.9	0	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.7	0	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.7	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.2	59	14.5	2.4	0.2	1.7
U3 - Toxicity prior Beta(0.15, 0.85)	26.2	56.5	12.3	3.1	0	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	21.3	57.7	15.5	3.7	0	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	24.5	55.5	15.2	3.1	0	1.7
U2 - Toxicity prior Beta(0.5, 0.5)	21.6	58	15.1	4	0.1	1.2
U3 - Toxicity prior Beta(0.5, 0.5)	26	56.1	13.3	3.5	0	1.1
U2 - Toxicity prior Beta(1, 1)	20.1	58	16	4.4	0.1	1.4
U3 - Toxicity prior Beta(1, 1)	23.9	56.6	13.6	4.4	0	1.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.6	12.7	6	2.5	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	12.5	5.7	2.4	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	12.5	6.2	2.7	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	8.8	12.3	5.9	2.6	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	8	12.4	6.3	2.9	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	12.3	6	2.8	0.4	
U2 - Toxicity prior Beta(1, 1)	7.7	12.2	6.4	3.2	0.4	
U3 - Toxicity prior Beta(1, 1)	8.3	12.2	6	3	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.6	1.7	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	4.5	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	4.5	1.5	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	4.4	1.5	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.2	1.4	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	4.1	1.4	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.9	3.8	1.3	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	1	3.6	1.2	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.3	4.1	0.1	0	0	74.5
U3 - Toxicity prior Beta(0.15, 0.85)	20	5.1	0	0	0	74.9
U2 - Toxicity prior Beta(0.3, 0.7)	22	4.4	0.3	0	0	73.3
U3 - Toxicity prior Beta(0.3, 0.7)	22.2	4.2	0.2	0	0	73.4
U2 - Toxicity prior Beta(0.5, 0.5)	23.3	4.6	0.1	0	0	72
U3 - Toxicity prior Beta(0.5, 0.5)	21.6	4.9	0.4	0	0	73.1
U2 - Toxicity prior Beta(1, 1)	22.1	4.2	0.1	0	0	73.6
U3 - Toxicity prior Beta(1, 1)	20.7	4.9	0	0	0	74.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.4	5.1	1	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	4.9	1	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.4	5.2	1.2	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	15.5	5	1.2	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.3	5.3	1.4	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.3	5.1	1.4	0.2	0	
U2 - Toxicity prior Beta(1, 1)	15.3	5.2	1.4	0.2	0	
U3 - Toxicity prior Beta(1, 1)	15.3	5	1.3	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.5	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.4	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.2	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.1	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	1.9	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	1.9	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.5	1.5	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.5	1.4	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	2.4	0.8	3.6	50.9	42.3	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	1.8	5.2	53.5	36	0
U2 - Toxicity prior Beta(0.3, 0.7)	2	0.9	3.7	50.6	42.8	0
U3 - Toxicity prior Beta(0.3, 0.7)	3.7	1.3	4.9	54.1	36	0
U2 - Toxicity prior Beta(0.5, 0.5)	2.2	1	3.8	50.8	42.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	3.6	1.6	4.6	53.2	37	0
U2 - Toxicity prior Beta(1, 1)	2.5	0.7	3.9	51.8	41.1	0
U3 - Toxicity prior Beta(1, 1)	3.6	1.6	5	52.9	36.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.9	11.8	8	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.9	12.6	7.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.1	3.9	11.9	8	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	4	12.4	7.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.9	12	8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.9	12.5	7.4	
U2 - Toxicity prior Beta(1, 1)	3.1	3.1	3.9	11.9	8	
U3 - Toxicity prior Beta(1, 1)	3.1	3.2	3.9	12.4	7.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1	2.5	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.4	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.6	31.2	33	5	0.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	38.5	31.5	25.9	4	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	30.5	29.8	33.3	6.1	0.2	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	36	30.3	29.1	4.3	0.2	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	28.1	30.3	35.4	5.8	0.3	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	35.7	28.9	30.5	4.6	0.2	0.1
U2 - Toxicity prior Beta(1, 1)	29.3	29.9	34.2	6.3	0.2	0.1
U3 - Toxicity prior Beta(1, 1)	35	30.8	28.3	5.5	0.3	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	9.2	8.7	3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	9.3	8.4	2.7	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.3	9	9	3.2	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	9.1	8.7	2.9	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8	8.9	9.2	3.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	9.1	8.8	3.1	0.5	
U2 - Toxicity prior Beta(1, 1)	7.9	8.7	9.3	3.5	0.6	
U3 - Toxicity prior Beta(1, 1)	8.6	8.9	8.8	3.2	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.7	1.9	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3.7	1.8	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.9	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3.5	1.8	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.4	1.8	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.4	1.7	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.2	1.6	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.2	1.5	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.1	36.7	29.8	5.5	0.1	2.8
U3 - Toxicity prior Beta(0.15, 0.85)	32.2	33.3	26.5	5	0.1	2.9
U2 - Toxicity prior Beta(0.3, 0.7)	25.3	35	30.2	6.6	0	2.9
U3 - Toxicity prior Beta(0.3, 0.7)	30	32.6	28	6.6	0	2.8
U2 - Toxicity prior Beta(0.5, 0.5)	26	32.4	32.4	6.5	0.1	2.6
U3 - Toxicity prior Beta(0.5, 0.5)	29.2	32.1	29.7	6.4	0	2.6
U2 - Toxicity prior Beta(1, 1)	23.7	33.9	32.2	7.3	0.2	2.7
U3 - Toxicity prior Beta(1, 1)	29.1	33.4	28.3	6.6	0.1	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	9	8.2	3.3	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	9.2	8	3.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.3	8.9	8.5	3.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.8	9.1	8.1	3.3	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7.8	8.8	8.8	3.8	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	8.9	8.4	3.6	0.5	
U2 - Toxicity prior Beta(1, 1)	7.8	8.6	8.7	4	0.7	
U3 - Toxicity prior Beta(1, 1)	8.3	8.6	8.5	3.8	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.7	1.2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.6	1.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.9	1.6	1.2	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3.9	1.5	1.1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.5	1.1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.6	1.4	1.1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.2	1.4	1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.2	1.4	1	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.7	27.7	44.1	9.9	0.6	1
U3 - Toxicity prior Beta(0.15, 0.85)	23.8	27.7	38.5	8.5	0.5	1
U2 - Toxicity prior Beta(0.3, 0.7)	16.5	26.7	44.3	11.1	0.6	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	22.6	27.7	39.7	8.8	0.4	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	16.7	25	46.1	10.7	1	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	21.2	25.7	42.2	9.5	0.9	0.5
U2 - Toxicity prior Beta(1, 1)	16.1	26	44.1	12.3	1.1	0.4
U3 - Toxicity prior Beta(1, 1)	19.5	26.8	42	10.4	0.9	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.5	8.1	10.1	4.3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	7	8.3	9.8	4	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	6.3	7.9	10.2	4.5	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	6.7	8.1	10	4.3	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	7.7	10.3	4.8	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	6.2	8.1	10.1	4.6	1	
U2 - Toxicity prior Beta(1, 1)	5.8	7.6	10.3	4.9	1.3	
U3 - Toxicity prior Beta(1, 1)	6.2	7.8	10.1	4.8	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.6	2.1	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.6	2	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.5	2	1.4	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.6	2	1.3	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	2	1.3	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1.9	1.3	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.1	1.9	1.2	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3.2	1.8	1.2	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.8	48.9	7.5	0.6	0	18.2
U3 - Toxicity prior Beta(0.15, 0.85)	26.9	47.4	6.1	0.4	0	19.2
U2 - Toxicity prior Beta(0.3, 0.7)	23.6	49	9.9	0.5	0	17
U3 - Toxicity prior Beta(0.3, 0.7)	26.5	48	8.2	0.2	0	17.1
U2 - Toxicity prior Beta(0.5, 0.5)	22.3	51.9	8.8	0.7	0.1	16.2
U3 - Toxicity prior Beta(0.5, 0.5)	23.6	51.6	7.3	0.7	0	16.8
U2 - Toxicity prior Beta(1, 1)	22	52.3	8.9	0.5	0	16.3
U3 - Toxicity prior Beta(1, 1)	24.4	50	8.5	0.7	0	16.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.1	11.8	4.1	1.3	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.5	11.8	3.9	1.2	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10.6	11.7	4.5	1.5	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11	11.7	4.3	1.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	10.2	12	4.7	1.6	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	12	4.5	1.5	0.2	
U2 - Toxicity prior Beta(1, 1)	10.1	11.6	4.9	1.7	0.3	
U3 - Toxicity prior Beta(1, 1)	10.4	11.9	4.6	1.6	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.4	1.2	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	4.4	1.1	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.2	1.1	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1	4.2	1.1	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3.8	1	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.8	0.9	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.8	3.1	0.9	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.8	3.2	0.9	0.3	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	62.6	30.9	3.9	0.3	0	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	66.9	27.9	2.8	0.1	0	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	60.2	31.8	5.5	0.3	0	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	65.9	28.6	3.2	0.1	0	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	59.4	31.8	6	0.4	0	2.4
U3 - Toxicity prior Beta(0.5, 0.5)	64.3	29.8	3.5	0.3	0	2.1
U2 - Toxicity prior Beta(1, 1)	59.9	31.9	5.5	0.5	0	2.2
U3 - Toxicity prior Beta(1, 1)	63.3	30	4	0.3	0	2.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.3	9	2.5	0.6	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.1	8.8	2.1	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16.7	9.3	2.9	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	17.6	9	2.4	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	16.1	9.5	3.1	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	9.3	2.6	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	16.1	9.4	3.3	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	17	9.2	2.6	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.1	0.9	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3	0.9	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	2.9	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.7	0.6	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.8	2.4	0.7	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.8	2.4	0.5	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.4	12.9	14.3	39.8	13.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	25.5	14.9	13.1	35.5	11	0
U2 - Toxicity prior Beta(0.3, 0.7)	19.3	12.1	13.4	40.7	14.5	0
U3 - Toxicity prior Beta(0.3, 0.7)	26.2	14.2	12.6	35.6	11.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	17.9	10.9	13.7	41.8	15.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	24.1	15.1	12.8	35.4	12.6	0
U2 - Toxicity prior Beta(1, 1)	17.1	11.2	14	41	16.7	0
U3 - Toxicity prior Beta(1, 1)	23.9	13.8	11.8	37.3	13.2	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	4.7	6.1	10.1	4.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	5	6.1	10.1	4.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	4.6	5.9	10.2	5.1	
U3 - Toxicity prior Beta(0.3, 0.7)	4.4	4.8	5.9	10.2	4.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.5	5.8	10.3	5.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	4.8	5.7	10.4	4.8	
U2 - Toxicity prior Beta(1, 1)	4.1	4.5	5.8	10.1	5.5	
U3 - Toxicity prior Beta(1, 1)	4.3	4.7	5.7	10.3	5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.8	2.4	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.8	2.4	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.7	2.4	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.7	2.3	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.7	2.4	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.6	2.3	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.7	2.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.7	2.2	1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	30.8	31.5	26.7	6.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	6.4	37.4	30.3	22.2	3.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	30.4	30.6	27.4	7.5	0
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	36.6	30	22.8	4.1	0
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	30.5	29.3	28	7.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	6.1	37.3	29.7	21.8	5.1	0
U2 - Toxicity prior Beta(1, 1)	5	32.3	28.2	27.1	7.4	0
U3 - Toxicity prior Beta(1, 1)	6.7	35.9	30.3	22.7	4.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	10.3	8.1	5.8	2	
U3 - Toxicity prior Beta(0.15, 0.85)	4	11.4	8.1	5.1	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	10.4	8	5.8	2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	11.3	7.9	5.3	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	10.3	7.9	5.9	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	11.2	7.9	5.3	1.7	
U2 - Toxicity prior Beta(1, 1)	3.8	10.4	7.7	5.9	2.2	
U3 - Toxicity prior Beta(1, 1)	3.9	11.1	7.8	5.4	1.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.6	1.3	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	1.2	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.6	1.3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	1.2	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.6	1.5	1.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.7	1.5	1.2	0.5	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	1.1	0.5	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14	13.1	33.4	27.3	12.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	19.8	16.9	32.6	23.7	7	0
U2 - Toxicity prior Beta(0.3, 0.7)	14.2	12.5	33.9	27.3	12.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	19.8	16.6	31.6	24.1	7.9	0
U2 - Toxicity prior Beta(0.5, 0.5)	12.8	13.2	33.4	26.8	13.8	0
U3 - Toxicity prior Beta(0.5, 0.5)	19.3	16.7	31.6	24.4	8	0
U2 - Toxicity prior Beta(1, 1)	13.2	12	32.7	28.1	14	0
U3 - Toxicity prior Beta(1, 1)	19.5	16	32	23.6	8.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4.1	8.3	8.5	5.5	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.5	8.6	8.5	4.8	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	4	8.3	8.6	5.7	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4.4	8.5	8.6	4.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	4	7.9	8.7	6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	4.2	8.4	8.8	5.1	
U2 - Toxicity prior Beta(1, 1)	3.4	3.9	8	8.7	6	
U3 - Toxicity prior Beta(1, 1)	3.5	4.1	8.3	8.7	5.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.8	2.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.9	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.8	2.9	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.7	2.9	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.9	2.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.7	2.9	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.8	2.8	1.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.4	5.1	13.7	41.6	29.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	14.3	7.2	13.7	40.3	24.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	4.5	12.8	40.6	31.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	13.2	7	12.4	40.4	27	0
U2 - Toxicity prior Beta(0.5, 0.5)	9.4	4	13.3	39	34.3	0
U3 - Toxicity prior Beta(0.5, 0.5)	13	7	12.3	40	27.7	0
U2 - Toxicity prior Beta(1, 1)	8.9	3.6	13.2	39.9	34.4	0
U3 - Toxicity prior Beta(1, 1)	13.1	6.8	11.7	39.7	28.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.6	5.9	9.9	7.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.8	5.9	10.3	6.5	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.6	5.8	9.9	7.4	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.7	5.7	10.3	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.6	5.7	9.8	7.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.7	5.6	10.4	6.9	
U2 - Toxicity prior Beta(1, 1)	3.4	3.6	5.4	9.7	7.9	
U3 - Toxicity prior Beta(1, 1)	3.4	3.7	5.4	10.2	7.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.5	2.7	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.5	2.7	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.7	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.7	1.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.6	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.6	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.3	13.9	43.4	21.5	5.6	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	21.1	16.1	40.4	18.8	3.4	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	16.4	12.6	42	22.2	6.7	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	21	15.9	39.3	19.5	4.2	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	16.1	13.4	41.8	20.8	7.7	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.5	15.3	40.3	18.9	4.8	0.2
U2 - Toxicity prior Beta(1, 1)	15.5	13.1	41.7	21.1	8.4	0.2
U3 - Toxicity prior Beta(1, 1)	21.4	14.2	40.3	18.4	5.5	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.1	6.2	9.2	6.5	2.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.7	6.4	9.2	6.2	2.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	6	9.2	6.7	3.3	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	6.1	9.1	6.4	2.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	5.8	9	6.8	3.6	
U3 - Toxicity prior Beta(0.5, 0.5)	5.1	6	9.1	6.6	3.2	
U2 - Toxicity prior Beta(1, 1)	4.7	5.6	8.9	6.9	3.9	
U3 - Toxicity prior Beta(1, 1)	5.1	5.8	9	6.7	3.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.4	2.1	2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	2	1.9	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	2	2.1	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.4	2	1.9	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.9	2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.9	1.9	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	3	1.8	2	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	3.1	1.8	1.8	0.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	40.2	33	13.9	3.7	0
U3 - Toxicity prior Beta(0.15, 0.85)	14	44.3	28.4	11.6	1.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	39.3	31.2	15.3	4.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	14.1	43.8	27.3	12.5	2.3	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	38.7	32.2	15.4	4.8	0
U3 - Toxicity prior Beta(0.5, 0.5)	13.7	43	27.9	12.7	2.7	0
U2 - Toxicity prior Beta(1, 1)	8.3	37.2	32	17.6	4.9	0
U3 - Toxicity prior Beta(1, 1)	11.9	43.9	28.3	13.4	2.5	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	10.9	8.6	4.6	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	11.9	8.2	4.1	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	10.6	8.5	4.8	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	11.6	8.3	4.3	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	10.6	8.5	4.8	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	11.7	8.2	4.3	1.4	
U2 - Toxicity prior Beta(1, 1)	4.2	10.2	8.5	5.2	1.9	
U3 - Toxicity prior Beta(1, 1)	4.3	11.6	8.2	4.4	1.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.9	1.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.1	1.9	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.8	1.3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.8	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.9	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3	1.8	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.9	1.8	1.3	0.3	
U3 - Toxicity prior Beta(1, 1)	0.2	3	1.8	1.1	0.3	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.8	43.1	17.7	6.7	3.3	5.4
U3 - Toxicity prior Beta(0.15, 0.85)	28.7	43.7	13.4	6.5	1.5	6.2
U2 - Toxicity prior Beta(0.3, 0.7)	21	45.2	17.7	8.8	2.9	4.4
U3 - Toxicity prior Beta(0.3, 0.7)	27.7	44.5	13.1	7.7	1.8	5.2
U2 - Toxicity prior Beta(0.5, 0.5)	20.7	45.2	17.3	8.9	4.1	3.8
U3 - Toxicity prior Beta(0.5, 0.5)	26	45.7	13.9	7.8	2.1	4.5
U2 - Toxicity prior Beta(1, 1)	19.5	43.1	17.7	10.4	4.6	4.7
U3 - Toxicity prior Beta(1, 1)	25.3	44.9	13.5	8.4	2.6	5.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	10	5.9	3.8	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	10	5.5	3.6	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	9.8	6.1	4	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	8.5	9.9	5.7	3.9	1.6	
U2 - Toxicity prior Beta(0.5, 0.5)	7.4	9.7	6.2	4.3	2.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8	9.6	6	4.2	1.9	
U2 - Toxicity prior Beta(1, 1)	7.2	9.4	6.2	4.6	2.2	
U3 - Toxicity prior Beta(1, 1)	7.9	9.5	5.9	4.4	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.5	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.3	1.5	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.3	1.5	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.2	1.4	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.9	1.4	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.9	1.4	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.7	3.5	1.3	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.8	3.5	1.3	1	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.7	54.2	15.5	4	0.8	2.8
U3 - Toxicity prior Beta(0.15, 0.85)	29.1	52.5	12.9	2.6	0.1	2.8
U2 - Toxicity prior Beta(0.3, 0.7)	24.1	51.3	16.9	4.2	1.2	2.3
U3 - Toxicity prior Beta(0.3, 0.7)	29.2	51.3	13	3.6	0.4	2.5
U2 - Toxicity prior Beta(0.5, 0.5)	22.9	51.7	17.4	4.7	1.2	2.1
U3 - Toxicity prior Beta(0.5, 0.5)	29.2	49.8	14.8	3.6	0.6	2
U2 - Toxicity prior Beta(1, 1)	22.8	50.1	18	4.9	1.4	2.8
U3 - Toxicity prior Beta(1, 1)	28.5	50.2	14.7	3.4	0.8	2.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.1	11.1	6	3.4	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	8.7	11.1	5.7	3.2	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	10.9	6.2	3.6	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	8.3	11	5.9	3.5	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	7.4	10.8	6.4	3.8	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8	10.7	6.1	3.7	1.4	
U2 - Toxicity prior Beta(1, 1)	7.1	10.6	6.5	4	1.6	
U3 - Toxicity prior Beta(1, 1)	7.8	10.6	6.1	3.8	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.8	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.6	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.6	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	4.5	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	4.3	1.4	1.1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1	4.2	1.4	1.1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.7	3.8	1.3	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.9	3.8	1.3	1	0.2	